
Introduction to Database

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Project

Deadline: Monday 09/08/2026 @ 23:59

[Total Mark is 20]

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Instructions:

- You must submit two separate copies (**one Word file and one PDF file**) using the Assignment Template on Blackboard via the allocated folder. These files **must not be in compressed format**.
- It is your responsibility to check and make sure that you have uploaded both the correct files.
- Zero mark will be given if you try to bypass the SafeAssign (e.g. misspell words, remove spaces between words, hide characters, use different character sets, **convert text into image** or languages other than English or any kind of manipulation).
- Email submission will not be accepted.
- You are advised to make your work clear and well-presented. This includes filling your information on the cover page.
- You must use this template, failing which will result in zero mark.
- You **MUST** show all your work, and text must not be converted into an image, unless specified otherwise by the question.
- Late submission will result in ZERO mark.
- The work should be your own, copying from students or other resources will result in ZERO mark.
- Use **Times New Roman** font for all your answers.

Project Instructions

- You can work on this project as a group (minimum 3 and maximum 4 students). Each group member must submit the project individually with all group member names mentioned on the cover page.
- This project **worth 20 marks** and will be distributed as in the following:
 - Database Design. (6 marks)
 - Database Implementation. (6 marks)
 - SQL Operations. (6 marks)
 - Project Reflection. (2 marks)
- Each leader in the group must submit one report about their chosen Project via the Blackboard (**Email submission will not be accepted and will be awarded ZERO marks**) containing the following:
 - A. Task 1 : database design
 - B. Task 2: database implementation
 - C. All requested queries/results
 - D. Screenshots from MySQL (or any other software you use) of **all the tables** after population and **query results**.
 - E. Reflection (300–500 words) describing the team's experience, challenges, design decisions, and AI usage (if applicable).
- Submit one **SQL file** containing the complete database implementation, including database creation, tables, data insertion, required queries, VIEW, and TRIGGER.
- Submit a **GitHub** (or GitLab) repository containing the complete project with regular commits and meaningful commit messages.
- Submit a **progress report** describing the completed work, key decisions, challenges, future plan, and contribution of each group member.
- Include **links** to the live report document, GitHub/GitLab repository, and Google Colab notebook (if applicable) in the **Project Artifacts** section of the final report.
- You are advised to make your work clear and well presented; marks may be reduced for poor presentation. This includes filling in your information on the cover page.
- You MUST show all your work, and text **must not** be converted into an image unless specified otherwise by the question.
- **Late submission will result in ZERO marks being awarded.**
- The work should be your own. Copying **from students or other resources, including AI software, will result in ZERO marks.**

*Learning**Outcome(s):**CLO1, CLO2, CLO3,**CLO4, CLO5*

Project Description

Design and Implementation of a Smart Clinic Database System

A private clinic currently manages its patient information manually, making it difficult to organize appointments, treatments, and payments. The clinic has decided to develop a database system to improve data organization and retrieval.

Your team is required to design, implement, and test a complete relational database using the concepts learned throughout this course. The project should demonstrate your understanding of database design, ER/EER modeling, relational mapping, SQL implementation, and advanced SQL queries. The project must be completed using MySQL.

Mid Project

1. Work Completed During This Period

During this phase, our team analyzed the clinic requirements and identified the main entities of the database, including Patients, Doctors, Appointments, Treatments, Medicines, and Payments, we designed the ER/EER model, defined the relationships between entities, assigned primary and foreign keys, created the relational schema, we also started implementing the database in MySQL by creating the tables with the required constraints and inserting sample records.

2. One Key Decision Made

One important design decision was to separate Appointments and Treatments into two related entities instead of storing all treatment information inside the appointment table, this follows the database design principles discussed in the course by reducing redundancy and improving data integrity.

3. One Problem Encountered

The main challenge was identifying the correct cardinality and foreign key relationships between patients, appointments, treatments, and payments, after reviewing the ER/EER model and testing the schema, we modified the relationships to ensure consistency and avoid data anomalies.

4. Plan for the Next Phase

Our next step is to complete the database implementation by inserting the remaining records, writing the required SQL queries, including SELECT, JOIN, Nested Queries, GROUP BY, UPDATE, and DELETE, creating the required VIEW and TRIGGER, finally, we will test the database and prepare the final report.

5. Team Members' Contributions

Reem Alshehri : Designed the ER/EER diagram and relational schema

Madaen Alshuqairan: Implemented the database tables in MySQL

Jana Almadhun: Inserted sample data and developed SQL queries

Riman Alhazme: Tested the database and prepared the documentation.

Task 1 – Database Design (6 Marks)

- Design a complete database for the clinic that includes at least six entities such as Patients, Doctors, Appointments, Treatments, Medicines, Payments, or any other appropriate entities.
- A complete ER/EER diagram.
- Primary and foreign keys.
- Appropriate relationships with cardinality.
- At least one specialization/generalization (EER feature).
- Any assumptions made during the design process.

Task 2 – Database Implementation (6 Marks)

Complete the following tasks using MySQL. Include screenshots of both your SQL code and the corresponding output (results) for each task:

- Create the database and all required tables.
- Use appropriate data types and constraints (PRIMARY KEY, FOREIGN KEY, NOT NULL, UNIQUE, etc.).
- Insert at least 5 records into each main table.

Task 3 – SQL Operations (6 Marks)

Complete the following tasks using MySQL. For each task, include screenshots of both your SQL code and the corresponding output (results), along with a brief description (one to two sentences) explaining the purpose of the query and the information it is intended to retrieve.

- Write SELECT statements.
- Write JOIN queries.
- Write nested queries.
- Use aggregate functions with GROUP BY.
- Write UPDATE and DELETE statements.
- Create one VIEW.
- Create one TRIGGER.

Task 4 – Reflection (2 Marks)

Write a **300–500-word reflection** on your experience while completing this project by answering the following questions. Your responses should be clear, concise, and supported with specific examples from your own work.

1. Describe one specific challenge or obstacle you encountered during this project and explain in detail how you identified and resolved it.
2. Explain why you chose your specific approach/method/model over at least one alternative you considered, and what tradeoffs informed that decision.
3. If you used any AI tools during this project (e.g., ChatGPT, Claude, Copilot), disclose which tool(s), for what specific purpose (e.g., debugging, grammar checking, brainstorming), and how you verified or modified the output.
4. What would you do differently if you were to redo this project with more time or resources?

Question 1

One of the main challenges we encountered was designing the relationships between Patients, Appointments, Treatments, and Payments while maintaining data integrity, at first we were unsure whether treatment information should be stored directly in the appointment table or in a separate table, after reviewing the ER/EER design principles and testing different relationships, we decided to create a separate Treatments entity linked to Appointments through foreign keys, this solution reduced data redundancy, improved flexibility, and maintained a normalized database structure.

Question 2

We chose a normalized relational database design because it minimizes data redundancy, improves data consistency, and maintains referential integrity through primary and foreign keys, the alternative approach was to combine several pieces of information into fewer tables to simplify the design, however, that approach would increase duplicate data and make future updates more difficult, although the

normalized design required more relationships and tables, it provides better scalability and follows the database design principles taught in the course.

Question 3

We used ChatGPT only for grammar checking and correcting minor language mistakes in the report, After reviewing the corrections, we ensured that the final text reflected our own work before including it in the report.

Question 4

If we had more time, we would improve the project by developing a graphical user interface connected to the database, adding more realistic datasets, and performing more comprehensive testing under different scenarios, we would also enhance the database by adding additional features such as appointment reminders, user authentication, and advanced reports, these improvements would make the system more practical and closer to a real clinic management system.

Deliverables

1. **Project Report (PDF)** – System description, ER/EER diagram, relational schema, SQL code, screenshots, and reflection.
2. **SQL Script (.sql)** – Complete SQL implementation of the project.
3. **GitHub Repository** – Repository link with the complete project and commit history.
4. **Mid-Project Progress Report** – Summary of progress, challenges, decisions, and team contributions.
5. **Project Artifacts**
 - Live report document (with edit history): [link]
 - GitHub/GitLab repository (with commit history): [link]
 - Google Colab notebook (if applicable): [link]